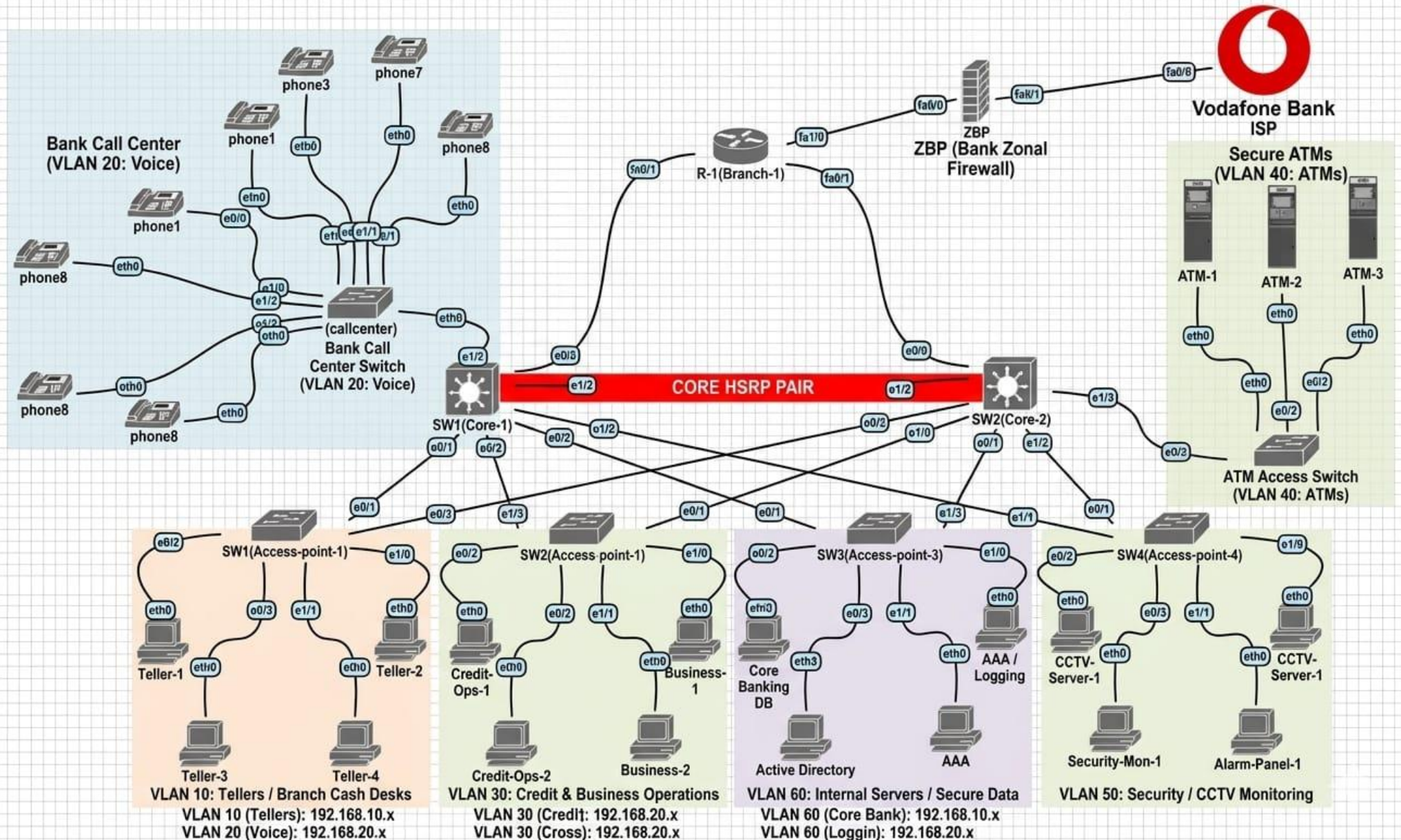


Banking network infrastructure



R-1 (Branch-1) - Edge Router

```
! =====  
! R-1 (Branch-1) - Edge Router  
! =====
```

```
enable  
configure terminal  
hostname R1-Branch1
```

```
! --- Interface to Vodafone ISP ---  
interface fa0/8  
description Link to Vodafone ISP  
ip address dhcp  
ip nat outside  
no shutdown
```

```
! --- Interface to ZBP Firewall ---  
interface fa1/0  
description Link to ZBP Firewall  
ip address 10.0.0.1 255.255.255.252  
ip nat inside  
no shutdown
```

```
! --- Interface to SW1 (Core-1) ---  
interface fa0/1
```

```
description Link to SW1-Core1
ip address 10.0.1.1 255.255.255.252
no shutdown
```

```
! --- Interface to SW2 (Core-2) ---
interface fa0/0
description Link to SW2-Core2
ip address 10.0.2.1 255.255.255.252
no shutdown
```

```
! --- OSPF ---
router ospf 1
router-id 1.1.1.1
network 10.0.0.0 0.0.0.3 area 0
network 10.0.1.0 0.0.0.3 area 0
network 10.0.2.0 0.0.0.3 area 0
default-information originate always
```

```
! --- NAT for Internet Access ---
ip nat inside source list 1 interface fa0/8 overload
access-list 1 permit 192.168.0.0 0.0.255.255
access-list 1 permit 10.0.0.0 0.0.255.255
```

```
! --- Default Route ---
ip route 0.0.0.0 0.0.0.0 fa0/8
```

! --- Security ---

ip domain-name vodafonebank.local

crypto key generate rsa modulus 2048

username admin privilege 15 secret Vodafone@Bank2024

line vty 0 4

login local

transport input ssh

enable secret Enable@VodafoneBank2024

! --- ACLs ---

access-list 101 deny ip 192.168.20.0 0.0.0.255 any

access-list 101 permit ip any any

end

write memory

ZBP (Bank Zonal Firewall)

```
! =====  
! ZBP - Bank Zonal Firewall  
! =====
```

```
enable  
configure terminal  
hostname ZBP-Firewall
```

```
! --- Outside Interface ---
```

```
interface fa1/1  
description Link to Vodafone ISP via R1  
nameif outside  
security-level 0  
ip address 10.0.0.2 255.255.255.252  
no shutdown
```

```
! --- Inside Interface ---
```

```
interface fa1/0  
description Link to R1-Branch1  
nameif inside  
security-level 100  
ip address 10.0.0.5 255.255.255.252  
no shutdown
```

```
! --- DMZ Interface (if needed) ---
```

```
interface fa0/0
 nameif dmz
 security-level 50
 ip address 192.168.100.1 255.255.255.0
 no shutdown
```

! --- NAT ---

```
nat (inside,outside) source dynamic any interface outside
nat (dmz,outside) source dynamic any interface outside
```

! --- Security Policies ---

! Block Voice VLAN from Internet

```
access-list OUTSIDE_IN extended deny ip 192.168.20.0 0.0.0.255 any
access-list OUTSIDE_IN extended permit icmp any any
access-list OUTSIDE_IN extended deny ip any any log
```

! Allow ATM traffic

```
access-list ATM_ACCESS extended permit tcp 192.168.40.0 0.0.0.255 any eq 443
access-list ATM_ACCESS extended permit tcp 192.168.40.0 0.0.0.255 any eq 80
```

! --- Access Groups ---

```
access-group OUTSIDE_IN in interface outside
```

! --- Routing ---

```
route outside 0.0.0.0 0.0.0.0 10.0.0.1
route inside 192.168.0.0 255.255.0.0 10.0.0.6
```

! --- SSH ---

ssh 10.0.0.0 255.255.255.252 inside

crypto key generate rsa modulus 2048

username admin password Vodafone@Bank2024 privilege 15

end

write memory

SW1 (Core-1) - HSRP Primary

```
! =====  
! SW1 (Core-1) - HSRP Primary  
! =====
```

```
enable  
configure terminal  
hostname SW1-Core1
```

```
! --- VTP ---  
vtp domain VODAFONE-BANK  
vtp mode server
```

```
! --- VLANs ---  
vlan 10  
    name Tellers-Cash-Desks  
vlan 20  
    name Voice-CallCenter  
vlan 30  
    name Credit-Business-Ops  
vlan 40  
    name Secure-ATMs  
vlan 50  
    name Security-CCTV  
vlan 60  
    name Internal-Servers
```



```
vlan 70
  name Logging
vlan 99
  name Management
```

```
! --- Interface to R1 ---
```

```
interface fa0/1
  description Uplink to R1-Branch1
  no switchport
  ip address 10.0.1.2 255.255.255.252
  no shutdown
```

```
! --- Interface to SW2 (HSRP Link) ---
```

```
interface e1/2
  description HSRP Link to SW2-Core2
  no switchport
  ip address 10.0.10.1 255.255.255.252
  no shutdown
```

```
! --- Trunk to Access Switches ---
```

```
interface range e0/1 - 3
  description Trunks to Access Switches
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30,40,50,60,70,99
  spanning-tree portfast trunk
```

no shutdown

! --- Trunk to Call Center Switch ---

interface e1/2

description Trunk to Call Center Switch

switchport trunk encapsulation dot1q

switchport mode trunk

switchport trunk allowed vlan 20,99

no shutdown

! --- SVI Interfaces with HSRP ---

! VLAN 10 - Tellers

interface vlan 10

ip address 192.168.10.2 255.255.255.0

standby 10 ip 192.168.10.1

standby 10 priority 110

standby 10 preempt

standby 10 track fa0/1

no shutdown

! VLAN 20 - Voice

interface vlan 20

ip address 192.168.20.2 255.255.255.0

standby 20 ip 192.168.20.1

standby 20 priority 110

standby 20 preempt

```
standby 20 track fa0/1  
no shutdown
```

! VLAN 30 - Credit & Business

```
interface vlan 30  
ip address 192.168.30.2 255.255.255.0  
standby 30 ip 192.168.30.1  
standby 30 priority 110  
standby 30 preempt  
standby 30 track fa0/1  
no shutdown
```

! VLAN 40 - ATMs

```
interface vlan 40  
ip address 192.168.40.2 255.255.255.0  
standby 40 ip 192.168.40.1  
standby 40 priority 110  
standby 40 preempt  
standby 40 track fa0/1  
no shutdown
```

! VLAN 50 - Security/CCTV

```
interface vlan 50  
ip address 192.168.50.2 255.255.255.0  
standby 50 ip 192.168.50.1  
standby 50 priority 110
```

```
standby 50 preempt
standby 50 track fa0/1
no shutdown
```

! VLAN 60 - Internal Servers

```
interface vlan 60
ip address 192.168.60.2 255.255.255.0
standby 60 ip 192.168.60.1
standby 60 priority 110
standby 60 preempt
standby 60 track fa0/1
no shutdown
```

! VLAN 70 - Logging

```
interface vlan 70
ip address 192.168.70.2 255.255.255.0
standby 70 ip 192.168.70.1
standby 70 priority 110
standby 70 preempt
standby 70 track fa0/1
no shutdown
```

! VLAN 99 - Management

```
interface vlan 99
ip address 192.168.99.2 255.255.255.0
standby 99 ip 192.168.99.1
```

```
standby 99 priority 110
standby 99 preempt
no shutdown
```

! --- Routing ---

```
ip routing
router ospf 1
router-id 2.2.2.2
network 10.0.1.0 0.0.0.3 area 0
network 10.0.10.0 0.0.0.3 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0
network 192.168.50.0 0.0.0.255 area 0
network 192.168.60.0 0.0.0.255 area 0
network 192.168.70.0 0.0.0.255 area 0
```

! --- Spanning Tree ---

```
spanning-tree mode rapid-pvst
spanning-tree vlan 10,20,30,40,50,60,70,99 root primary
```

! --- DHCP Helper ---

```
interface vlan 10
ip helper-address 192.168.60.10
interface vlan 20
```

```
ip helper-address 192.168.60.10
interface vlan 30
ip helper-address 192.168.60.10
interface vlan 40
ip helper-address 192.168.60.10
interface vlan 50
ip helper-address 192.168.60.10
```

! --- QoS for Voice ---

```
mls qos
interface vlan 20
mls qos trust cos
priority-queue out
```

! --- Security ---

```
ip domain-name vodafonebank.local
crypto key generate rsa modulus 2048
username admin privilege 15 secret Vodafone@Bank2024
line vty 0 15
login local
transport input ssh
enable secret Enable@VodafoneBank2024
```

! --- SNMP ---

```
snmp-server community VodafoneBankSNMP RO
snmp-server location "Vodafone Bank Core-1"
```

end

write memory

SW2 (Core-2) - HSRP Standby

```
! =====  
! SW2 (Core-2) - HSRP Standby  
! =====
```

```
enable  
configure terminal  
hostname SW2-Core2
```

```
! --- VTP ---  
vtp domain VODAFONE-BANK  
vtp mode client
```

```
! --- Interface to R1 ---  
interface fa0/0  
description Uplink to R1-Branch1  
no switchport  
ip address 10.0.2.2 255.255.255.252  
no shutdown
```

```
! --- Interface to SW1 (HSRP Link) ---  
interface e1/2  
description HSRP Link to SW1-Core1  
no switchport  
ip address 10.0.10.2 255.255.255.252  
no shutdown
```


! --- Trunk to Access Switches ---

```
interface range e0/1 - 3
description Trunks to Access Switches
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,70,99
spanning-tree portfast trunk
no shutdown
```

! --- Trunk to ATM Switch ---

```
interface e1/3
description Trunk to ATM Access Switch
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 40,99
no shutdown
```

! --- SVI Interfaces with HSRP (Lower Priority) ---

! VLAN 10 - Tellers

```
interface vlan 10
ip address 192.168.10.3 255.255.255.0
standby 10 ip 192.168.10.1
standby 10 priority 100
standby 10 preempt
standby 10 track fa0/0
```

no shutdown

! VLAN 20 - Voice

interface vlan 20

ip address 192.168.20.3 255.255.255.0

standby 20 ip 192.168.20.1

standby 20 priority 100

standby 20 preempt

standby 20 track fa0/0

no shutdown

! VLAN 30 - Credit & Business

interface vlan 30

ip address 192.168.30.3 255.255.255.0

standby 30 ip 192.168.30.1

standby 30 priority 100

standby 30 preempt

standby 30 track fa0/0

no shutdown

! VLAN 40 - ATMs

interface vlan 40

ip address 192.168.40.3 255.255.255.0

standby 40 ip 192.168.40.1

standby 40 priority 100

standby 40 preempt

```
standby 40 track fa0/0  
no shutdown
```

! VLAN 50 - Security/CCTV

```
interface vlan 50  
ip address 192.168.50.3 255.255.255.0  
standby 50 ip 192.168.50.1  
standby 50 priority 100  
standby 50 preempt  
standby 50 track fa0/0  
no shutdown
```

! VLAN 60 - Internal Servers

```
interface vlan 60  
ip address 192.168.60.3 255.255.255.0  
standby 60 ip 192.168.60.1  
standby 60 priority 100  
standby 60 preempt  
standby 60 track fa0/0  
no shutdown
```

! VLAN 70 - Logging

```
interface vlan 70  
ip address 192.168.70.3 255.255.255.0  
standby 70 ip 192.168.70.1  
standby 70 priority 100
```

```
standby 70 preempt
standby 70 track fa0/0
no shutdown
```

! VLAN 99 - Management

```
interface vlan 99
ip address 192.168.99.3 255.255.255.0
standby 99 ip 192.168.99.1
standby 99 priority 100
standby 99 preempt
no shutdown
```

! --- Routing ---

```
ip routing
router ospf 1
router-id 3.3.3.3
network 10.0.2.0 0.0.0.3 area 0
network 10.0.10.0 0.0.0.3 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0
network 192.168.50.0 0.0.0.255 area 0
network 192.168.60.0 0.0.0.255 area 0
network 192.168.70.0 0.0.0.255 area 0
```

! --- Spanning Tree ---

spanning-tree mode rapid-pvst

spanning-tree vlan 10,20,30,40,50,60,70,99 root secondary

! --- DHCP Helper ---

interface vlan 10

ip helper-address 192.168.60.10

interface vlan 20

ip helper-address 192.168.60.10

interface vlan 30

ip helper-address 192.168.60.10

interface vlan 40

ip helper-address 192.168.60.10

interface vlan 50

ip helper-address 192.168.60.10

! --- QoS for Voice ---

mls qos

interface vlan 20

mls qos trust cos

priority-queue out

! --- Security ---

ip domain-name vodafonebank.local

crypto key generate rsa modulus 2048

username admin privilege 15 secret Vodafone@Bank2024

```
line vty 0 15
 login local
 transport input ssh
 enable secret Enable@VodafoneBank2024
```

```
! --- SNMP ---
snmp-server community VodafoneBankSNMP RO
snmp-server location "Vodafone Bank Core-2"
```

```
end
write memory
```

Call Center Switch (VLAN 20: Voice)

```
! =====  
! Call Center Switch - Voice VLAN 20  
! =====
```

```
enable  
configure terminal  
hostname CallCenter-SW
```

```
! --- VTP ---  
vtp domain VODAFONE-BANK  
vtp mode client
```

```
! --- VLAN ---  
vlan 20  
name Voice-CallCenter
```

```
! --- Uplink to Core ---  
interface e1/2  
description Uplink to SW1-Core1  
switchport trunk encapsulation dot1q  
switchport mode trunk  
switchport trunk allowed vlan 20,99  
no shutdown
```

```
! --- Voice Ports for IP Phones ---
```

```
interface range e0/0 - 3, e1/0 - 1
description IP Phones
switchport mode access
switchport access vlan 20
switchport voice vlan 20
spanning-tree portfast
no shutdown
```

! --- QoS ---

```
mls qos
interface range e0/0 - 3, e1/0 - 1
mls qos trust cos
priority-queue out
```

! --- Management ---

```
interface vlan 99
ip address 192.168.99.10 255.255.255.0
no shutdown
```

```
ip default-gateway 192.168.99.1
```

! --- Security ---

```
ip domain-name vodafonebank.local
crypto key generate rsa modulus 2048
username admin privilege 15 secret Vodafone@Bank2024
line vty 0 4
```


login local

transport input ssh

end

write memory

SW1 (Access-point-1) - Tellers VLAN 10

```
! =====  
! SW1-Access-1 - Tellers / Cash Desks (VLAN 10)  
! =====
```

```
enable  
configure terminal  
hostname SW1-Access-Tellers
```

```
! --- VTP ---  
vtp domain VODAFONE-BANK  
vtp mode client
```

```
! --- VLANs ---  
vlan 10  
    name Tellers-Cash-Desks  
vlan 20  
    name Voice
```

```
! --- Uplink to Core ---  
interface e0/1  
    description Uplink to SW1-Core1  
    switchport trunk encapsulation dot1q  
    switchport mode trunk  
    switchport trunk allowed vlan 10,20,99  
    no shutdown
```

```
interface e0/3
description Uplink to SW2-Core2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,99
no shutdown
```

! --- Access Ports - Teller PCs ---

```
interface range e0/0, e1/1
description Teller PCs
switchport mode access
switchport access vlan 10
switchport port-security
switchport port-security maximum 2
switchport port-security violation restrict
switchport port-security mac-address sticky
spanning-tree portfast
no shutdown
```

! --- Voice Ports ---

```
interface range e0/2, e1/0
description IP Phones for Tellers
switchport mode access
switchport access vlan 10
switchport voice vlan 20
```

```
spanning-tree portfast  
no shutdown
```

```
! --- Management ---  
interface vlan 99  
ip address 192.168.99.11 255.255.255.0  
no shutdown
```

```
ip default-gateway 192.168.99.1
```

```
! --- Security ---  
ip domain-name vodafonebank.local  
crypto key generate rsa modulus 2048  
username admin privilege 15 secret Vodafone@Bank2024  
line vty 0 4  
login local  
transport input ssh
```

```
end  
write memory
```

SW2 (Access-point-2) - Credit & Business VLAN 30

```
! =====  
! SW2-Access-2 - Credit & Business Operations (VLAN 30)  
! =====
```

```
enable  
configure terminal  
hostname SW2-Access-Credit
```

```
! --- VTP ---  
vtp domain VODAFONE-BANK  
vtp mode client
```

```
! --- VLAN ---  
vlan 30  
name Credit-Business-Ops
```

```
! --- Uplinks to Core ---  
interface e0/1  
description Uplink to SW1-Core1  
switchport trunk encapsulation dot1q  
switchport mode trunk  
switchport trunk allowed vlan 30,99  
no shutdown
```

```
interface e1/0
```

```
description Uplink to SW2-Core2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 30,99
no shutdown
```

! --- Access Ports ---

```
interface range e0/0, e0/2, e1/1
description Credit & Business PCs
switchport mode access
switchport access vlan 30
switchport port-security
switchport port-security maximum 2
switchport port-security violation restrict
spanning-tree portfast
no shutdown
```

! --- Management ---

```
interface vlan 99
ip address 192.168.99.12 255.255.255.0
no shutdown
```

```
ip default-gateway 192.168.99.1
```

! --- Security ---

```
ip domain-name vodafonebank.local
```

```
crypto key generate rsa modulus 2048
username admin privilege 15 secret Vodafone@Bank2024
line vty 0 4
  login local
  transport input ssh

end
write memory
```

SW3 (Access-point-3) - Internal Servers VLAN 60

```
! =====  
! SW3-Access-3 - Internal Servers / Secure Data (VLAN 60)  
! =====
```

```
enable  
configure terminal  
hostname SW3-Access-Servers
```

```
! --- VTP ---  
vtp domain VODAFONE-BANK  
vtp mode client
```

```
! --- VLANs ---  
vlan 60  
    name Internal-Servers  
vlan 70  
    name Logging
```

```
! --- Uplinks to Core ---  
interface e0/1  
    description Uplink to SW1-Core1  
    switchport trunk encapsulation dot1q  
    switchport mode trunk  
    switchport trunk allowed vlan 60,70,99  
    no shutdown
```



```
interface e1/0
description Uplink to SW2-Core2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 60,70,99
no shutdown
```

! --- Server Ports ---

```
interface e0/0
description Core Banking DB
switchport mode access
switchport access vlan 60
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown
spanning-tree portfast
no shutdown
```

```
interface e0/2
description Active Directory
switchport mode access
switchport access vlan 60
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown
```

```
spanning-tree portfast
no shutdown
```

```
interface e1/1
description AAA / Logging Server
switchport mode access
switchport access vlan 70
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown
spanning-tree portfast
no shutdown
```

! --- Management ---

```
interface vlan 99
ip address 192.168.99.13 255.255.255.0
no shutdown
```

```
ip default-gateway 192.168.99.1
```

! --- Security ---

```
ip domain-name vodafonebank.local
crypto key generate rsa modulus 2048
username admin privilege 15 secret Vodafone@Bank2024
line vty 0 4
login local
```

transport input ssh

end

write memory

SW4 (Access-point-4) - Security / CCTV VLAN 50

```
! =====  
! SW4-Access-4 - Security / CCTV Monitoring (VLAN 50)  
! =====
```

```
enable  
configure terminal  
hostname SW4-Access-Security
```

```
! --- VTP ---  
vtp domain VODAFONE-BANK  
vtp mode client
```

```
! --- VLAN ---  
vlan 50  
name Security-CCTV
```

```
! --- Uplinks to Core ---  
interface e0/1  
description Uplink to SW1-Core1  
switchport trunk encapsulation dot1q  
switchport mode trunk  
switchport trunk allowed vlan 50,99  
no shutdown
```

```
interface e1/0
```

```
description Uplink to SW2-Core2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 50,99
no shutdown
```

! --- CCTV Server Ports ---

```
interface e0/0
description CCTV-Server-1
switchport mode access
switchport access vlan 50
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown
spanning-tree portfast
no shutdown
```

```
interface e0/2
description Security-Mon-1
switchport mode access
switchport access vlan 50
switchport port-security
switchport port-security maximum 1
spanning-tree portfast
no shutdown
```

```
interface e1/1
description Alarm-Panel-1
switchport mode access
switchport access vlan 50
switchport port-security
switchport port-security maximum 1
spanning-tree portfast
no shutdown
```

! --- Management ---

```
interface vlan 99
ip address 192.168.99.14 255.255.255.0
no shutdown
```

```
ip default-gateway 192.168.99.1
```

! --- Security ---

```
ip domain-name vodafonebank.local
crypto key generate rsa modulus 2048
username admin privilege 15 secret Vodafone@Bank2024
line vty 0 4
login local
transport input ssh
end
write memory
```

ATM Access Switch (VLAN 40)

```
! =====  
! ATM Access Switch - Secure ATMs (VLAN 40)  
! =====
```

```
enable  
configure terminal  
hostname ATM-Access-SW
```

```
! --- VTP ---  
vtp domain VODAFONE-BANK  
vtp mode client
```

```
! --- VLAN ---  
vlan 40  
name Secure-ATMs
```

```
! --- Uplink to Core ---  
interface e0/3  
description Uplink to SW2-Core2  
switchport trunk encapsulation dot1q  
switchport mode trunk  
switchport trunk allowed vlan 40,99  
no shutdown
```

```
! --- ATM Ports ---
```

```
interface range e0/0 - 2
description ATM Machines
switchport mode access
switchport access vlan 40
switchport port-security
switchport port-security maximum 1
switchport port-security violation shutdown
switchport port-security mac-address sticky
spanning-tree portfast
no shutdown
```

! --- Security: Isolate ATMs ---

```
interface range e0/0 - 2
switchport protected
```

! --- Management ---

```
interface vlan 99
ip address 192.168.99.15 255.255.255.0
no shutdown
```

```
ip default-gateway 192.168.99.1
```

! --- ACL for ATM VLAN ---

```
access-list 140 permit tcp any any eq 443
access-list 140 permit tcp any any eq 80
access-list 140 permit udp any any eq 53
```



```
access-list 140 deny ip any any log
```

```
! --- Security ---
```

```
ip domain-name vodafonebank.local
```

```
crypto key generate rsa modulus 2048
```

```
username admin privilege 15 secret Vodafone@Bank2024
```

```
line vty 0 4
```

```
login local
```

```
transport input ssh
```

```
end
```

```
write memory
```

DHCP Server Configuration (on Internal Server)

```
! =====  
! DHCP Server on Core Switch (or Windows/Linux Server)  
! =====
```

```
! --- On SW1-Core1 ---
```

```
ip dhcp pool VLAN10-TELLERS  
network 192.168.10.0 255.255.255.0  
default-router 192.168.10.1  
dns-server 192.168.60.10, 8.8.8.8  
domain-name vodafonebank.local  
lease 8
```

```
ip dhcp pool VLAN20-VOICE  
network 192.168.20.0 255.255.255.0  
default-router 192.168.20.1  
option 150 ip 192.168.60.20  
dns-server 192.168.60.10  
domain-name vodafonebank.local  
lease 8
```

```
ip dhcp pool VLAN30-CREDIT  
network 192.168.30.0 255.255.255.0  
default-router 192.168.30.1  
dns-server 192.168.60.10, 8.8.8.8  
domain-name vodafonebank.local
```

lease 8

```
ip dhcp pool VLAN40-ATMS
network 192.168.40.0 255.255.255.0
default-router 192.168.40.1
dns-server 192.168.60.10
domain-name vodafonebank.local
lease 8
```

```
ip dhcp pool VLAN50-SECURITY
network 192.168.50.0 255.255.255.0
default-router 192.168.50.1
dns-server 192.168.60.10
domain-name vodafonebank.local
lease 8
```

! --- Exclude Server IPs ---

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp excluded-address 192.168.30.1 192.168.30.10
ip dhcp excluded-address 192.168.40.1 192.168.40.10
ip dhcp excluded-address 192.168.50.1 192.168.50.10
ip dhcp excluded-address 192.168.60.1 192.168.60.20
ip dhcp excluded-address 192.168.70.1 192.168.70.10
```

◆ Security Policies Summary

cisco

```
! =====  
! Security Policies - Vodafone Bank  
! =====
```

! 1. Inter-VLAN ACLs

! Block Voice VLAN from accessing sensitive data

```
access-list 120 deny ip 192.168.20.0 0.0.0.255 192.168.60.0 0.0.0.255  
access-list 120 deny ip 192.168.20.0 0.0.0.255 192.168.70.0 0.0.0.255  
access-list 120 permit ip any any
```

! 2. ATM Isolation

! ATMs can only access banking servers

```
access-list 140 permit tcp 192.168.40.0 0.0.0.255 host 192.168.60.10 eq 443  
access-list 140 permit tcp 192.168.40.0 0.0.0.255 host 192.168.60.10 eq 80  
access-list 140 deny ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255  
access-list 140 deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255  
access-list 140 deny ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255  
access-list 140 permit ip any any
```

! 3. CCTV Security

! CCTV can access all but cannot be accessed from outside

```
access-list 150 permit ip 192.168.50.0 0.0.0.255 any  
access-list 150 deny ip any 192.168.50.0 0.0.0.255
```

! 4. Management VLAN Isolation

```
access-list 199 permit ip 192.168.99.0 0.0.0.255 192.168.99.0 0.0.0.255
```

```
access-list 199 deny ip any 192.168.99.0 0.0.0.255  
access-list 199 permit ip any any
```

! 5. Apply ACLs

```
interface vlan 20
```

```
ip access-group 120 in
```

```
interface vlan 40
```

```
ip access-group 140 in
```

```
interface vlan 50
```

```
ip access-group 150 in
```

```
interface vlan 99
```

```
ip access-group 199 in
```